

Towards Validated SWIR Hyperspectral Methods for Small Microplastic (≤ 100 μm) Detection in Water Filtration Samples: Challenges and Perspectives

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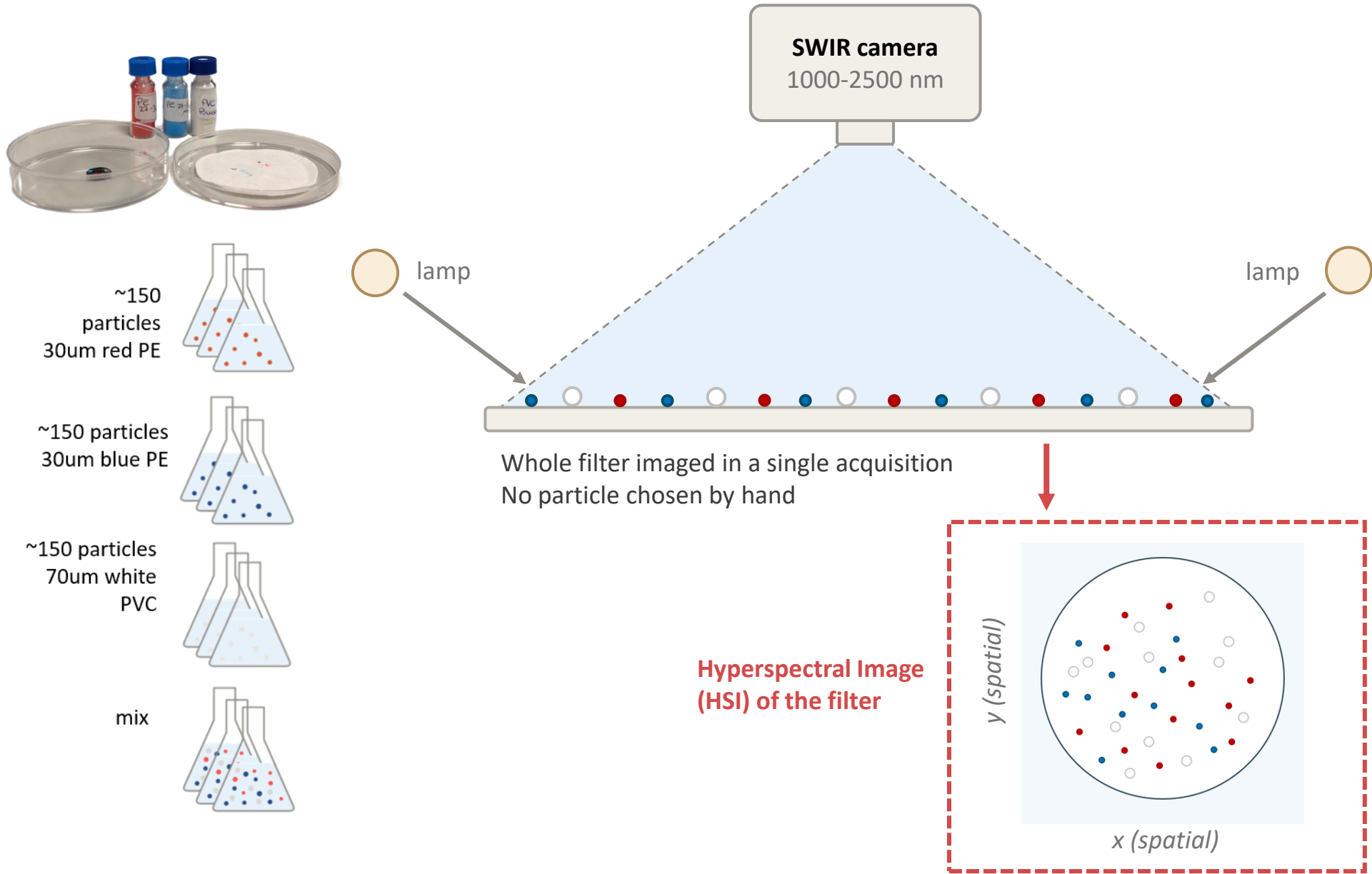
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** Presenting author



What was done: SWIR Hyperspectral Imaging



What was done: Pixel-by-pixel Comparison with Ref

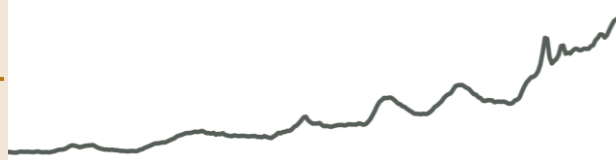
HSI Image, N pixels

One image per wavelength

λ (wavelength)

One pixel, one HSI spectrum

Average of photons detected at a pixel

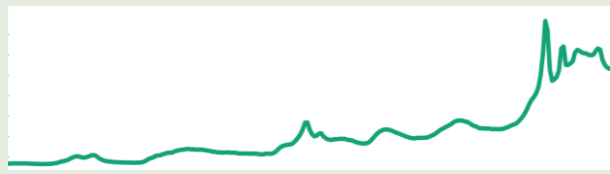


The process

Comparison with:

Reference HSI spectrum

E.g., pure plastic, non-plastic, etc.

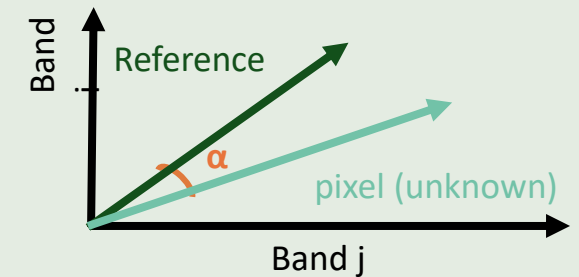


Spectral Angle Mapper (SAM)

Target method for anomaly detection

Each pixel is assigned:

- α : Similarity angle between a pixel spectrum and a reference. A small α means the more similar.
- **Cosine α** : 1 means similar to the reference

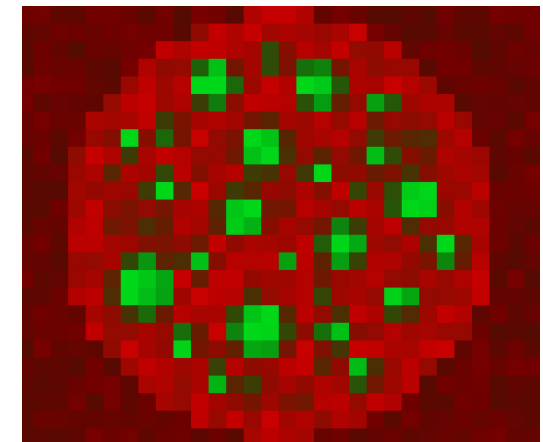


The dream

XX number of Blue PE
YY number of Red PE
ZZ number of White PVC

Image segmentation and particle counting:

- Cutoff for α or **Cosine α**
- 1 particle = n pixels



Most similar to plastic

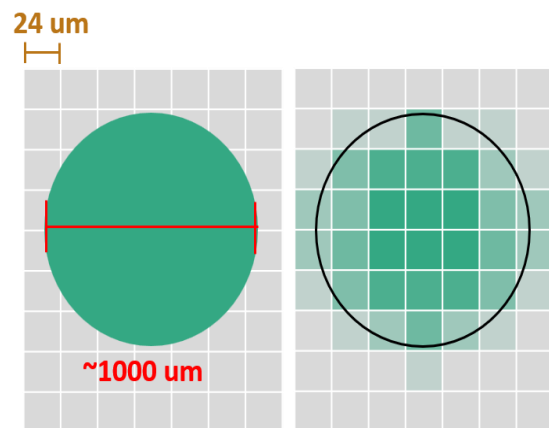
Most dissimilar to plastic

Are we counting the correct pixels?

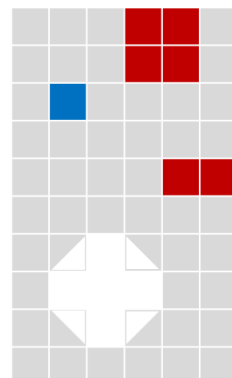
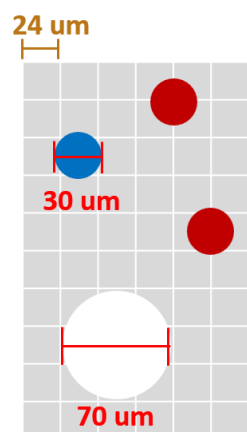
Challenge 1: ???

A **pixel** is a **sampling grid**: a small square that can be two things at once.

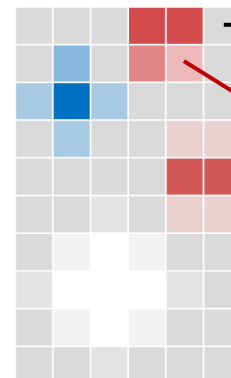
Big MPs ($\gg$ pixel size):



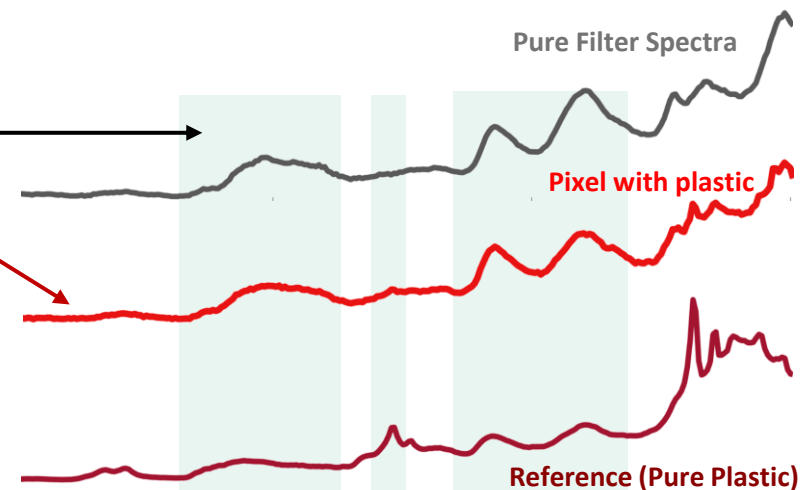
Small MPs ($\cong$ pixel size):



What we wish we
had: clean pixel



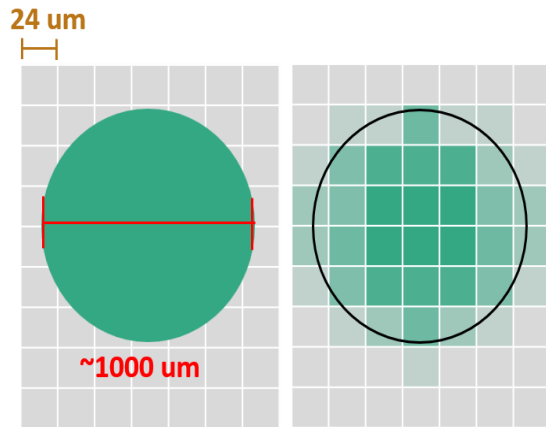
What we get: pixel-
mixing, scattering, etc



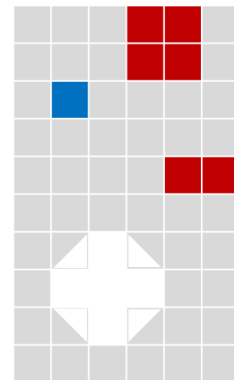
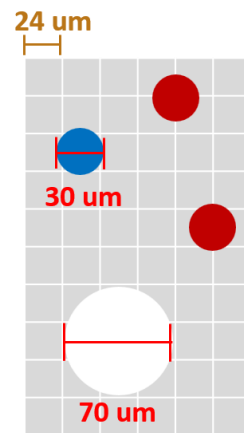
Challenge 1: A pixel is rarely a pure spectrum of a particle

A pixel is a sampling grid: a small square that can be two things at once.

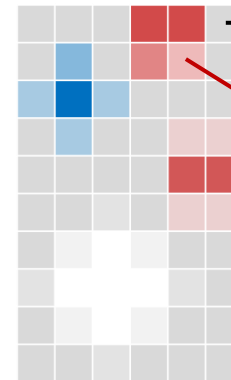
Big MPs ($\gg$ pixel size):



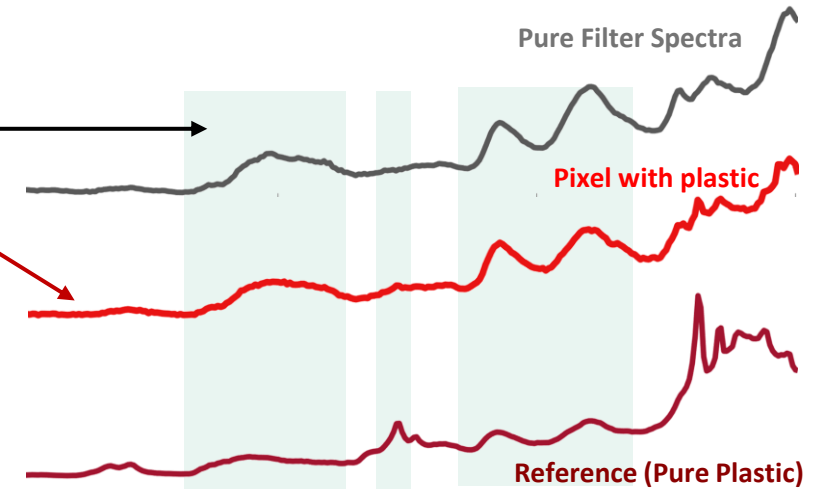
Small MPs ($\cong$ pixel size):



What we wish we had: clean pixel



What we get: pixel-mixing, scattering, etc



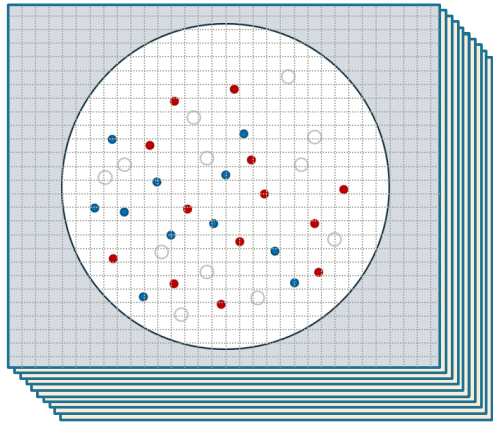
- Core pixels: near-pure spectra of the object
- Rim pixels: mixtures (partial coverage, translucency, scattering, etc)

- Small and thin
→ translucency + scattering from the sides
- Every pixel is a mixture:
→ particle + background + scattering
- Pixel WITH a plastic particle, not OF....

Perspective: Addressing Pixel-Mixing and Scattering

- Correct Scattering → **Preprocessing**
- Accept Mixing as the normal state → Include the mixture in the model (e.g, **Match the mixture**)

HSI Image, N pixels



1

Spectral
Preprocessing

Reference HSI spectrum

2

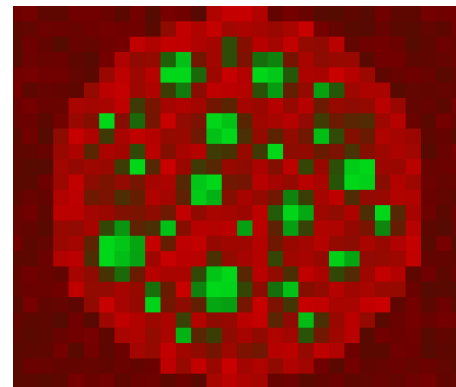
Pure plastic

Plastic on a filter

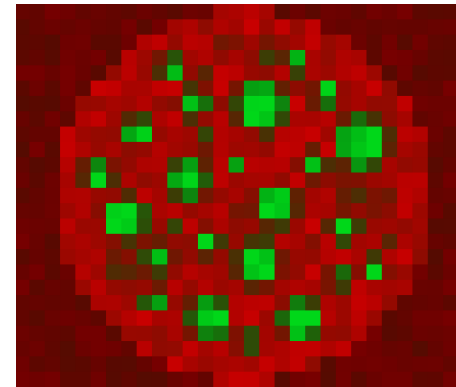
3

Spectral Angle Mapper (SAM)

Target method for anomaly detection



SAM vs pure plastic



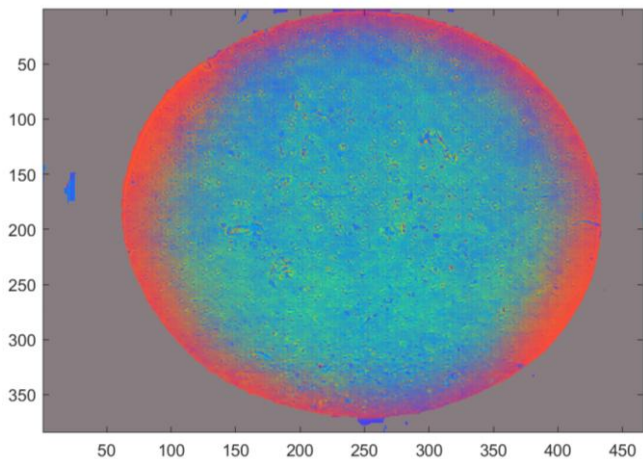
SAM vs plastic on a filter

Image
segmentation
and particle
counting:

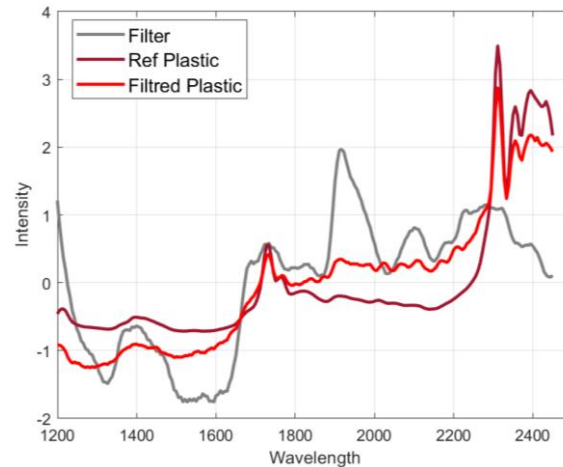
Cutoff set to segment
plastic and non-plastic

4

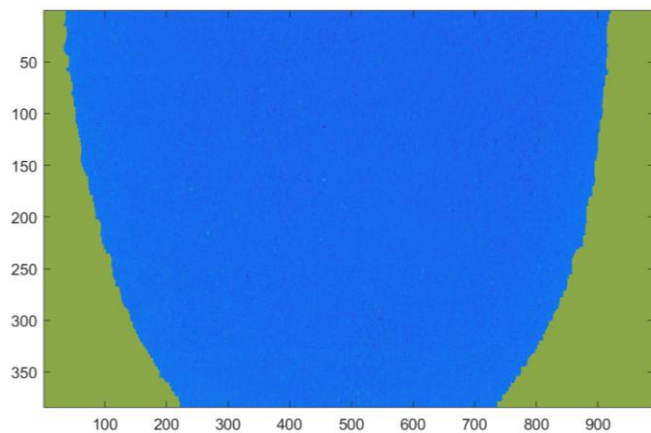
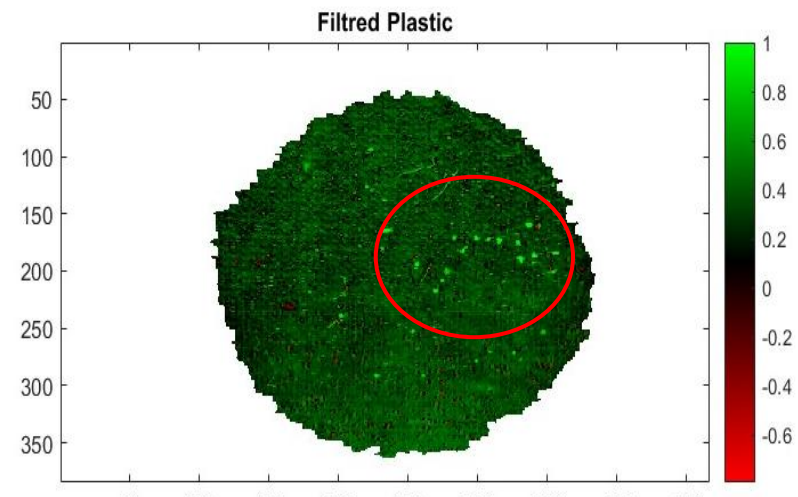
Challenge 2: ??



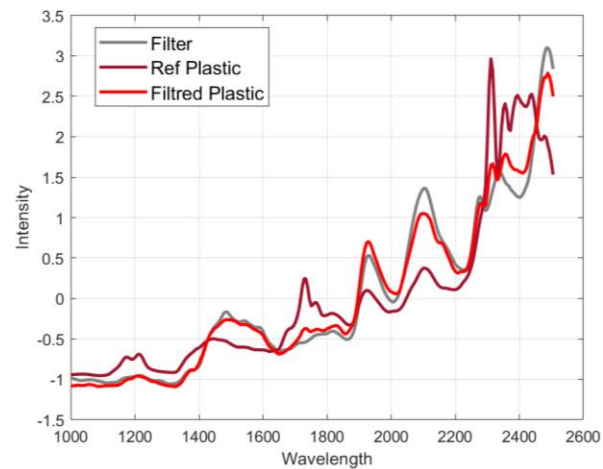
SNV →



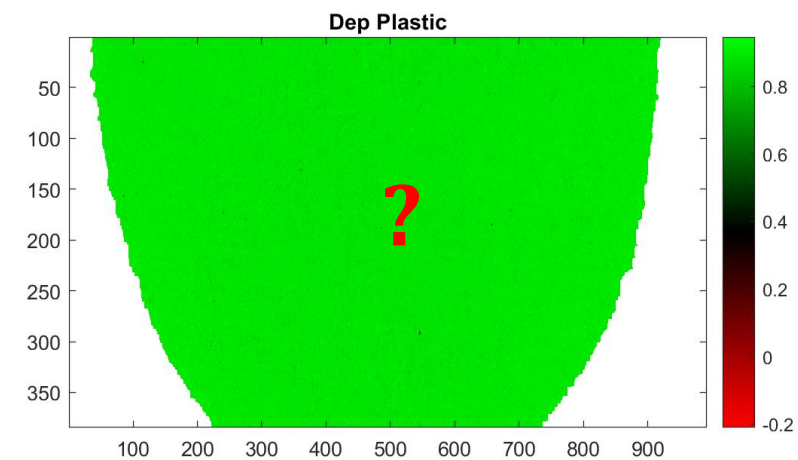
SAM →



SNV →

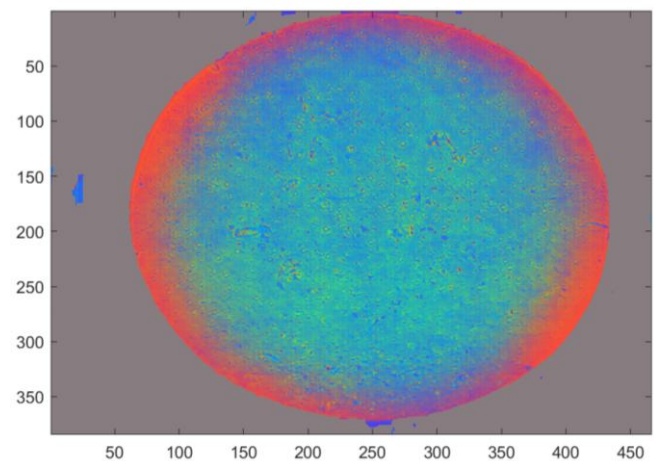


SAM →

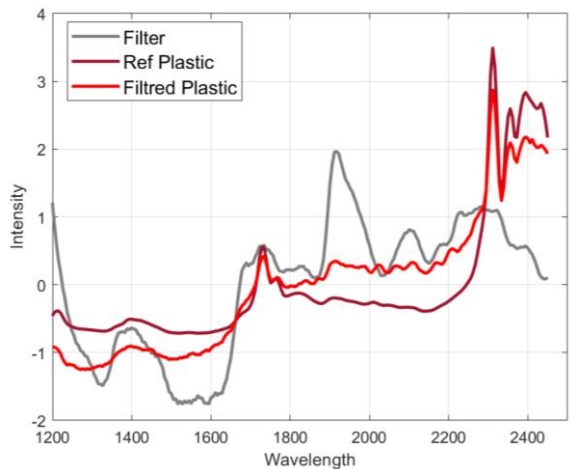


Cosine Maps against reference

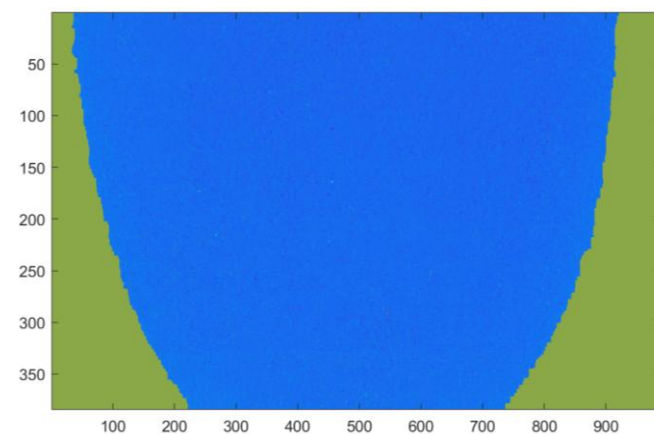
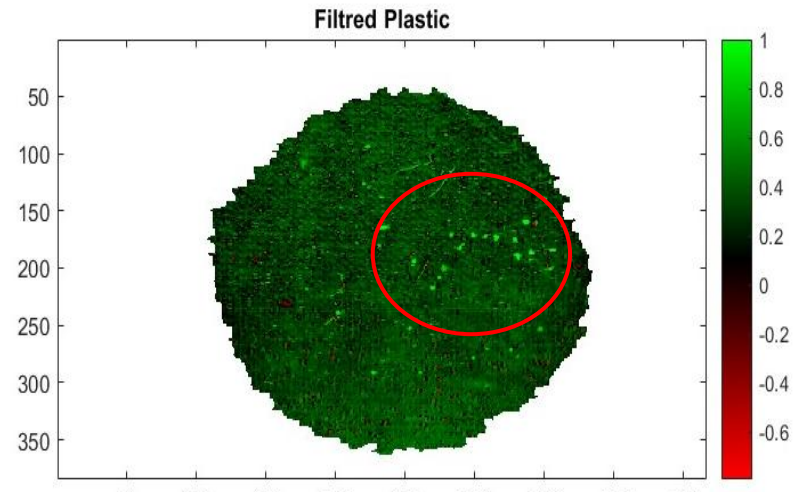
Challenge 2: No Single Preprocessing Recipe



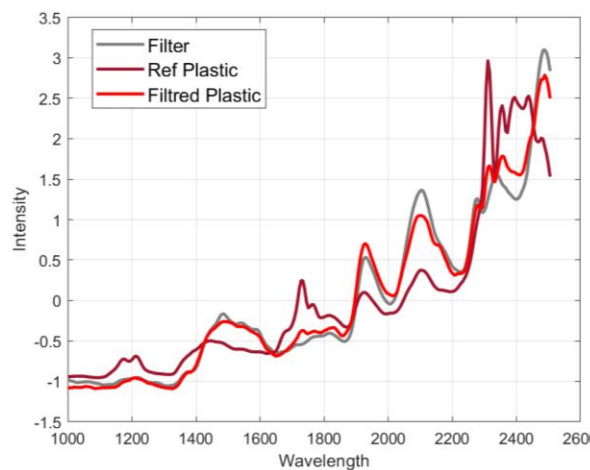
SNV



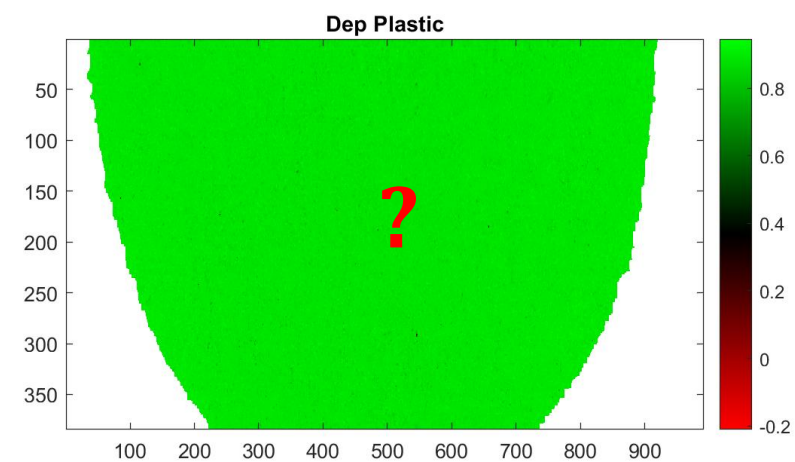
SAM



SNV



SAM



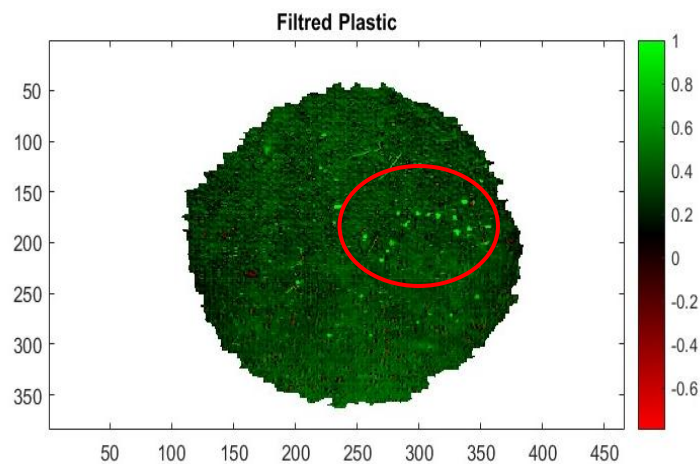
Cosine Maps against reference

Perspective: Addressing Preprocessing Challenges

- Tailor preprocessing based on needs

Case 1:

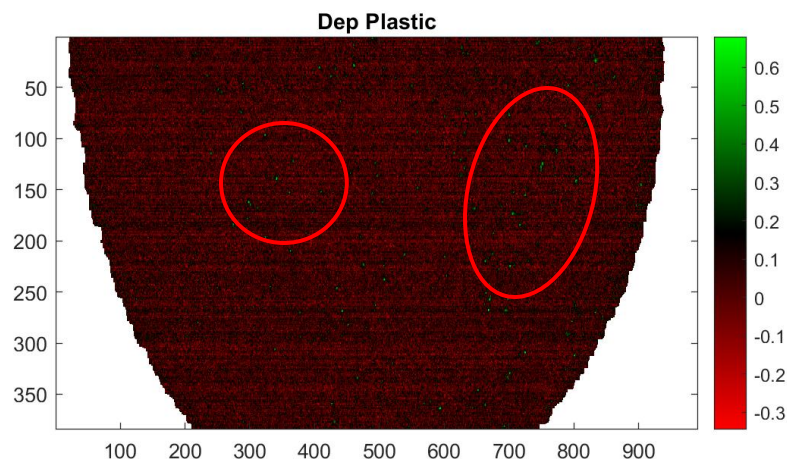
Plastic and Filter are very different
(PE on Silicon Filter)



SNV works

Case 2:

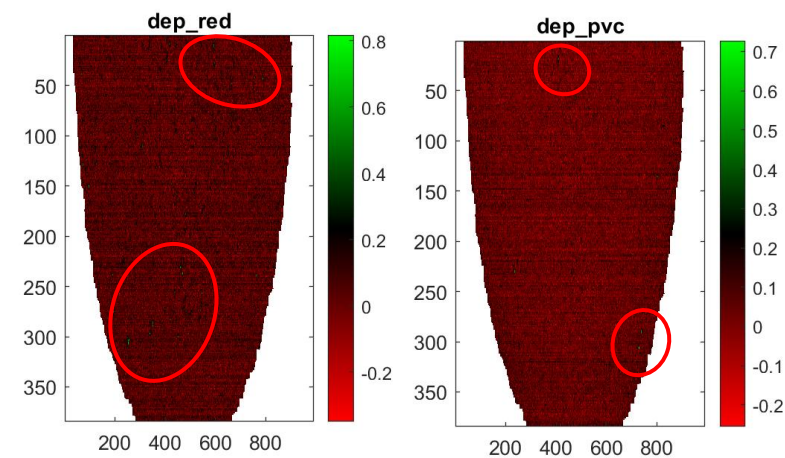
Plastic and Filter are similar + more...
(PE on Cellulose Filter)



2D+SNV

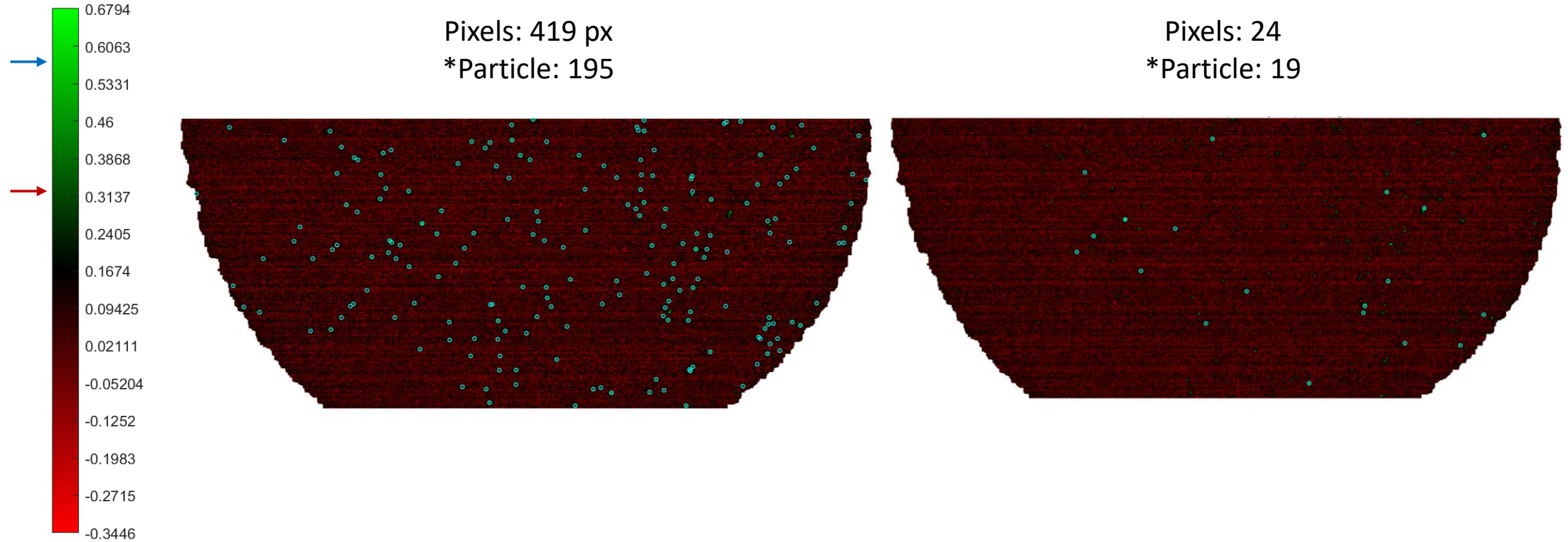
Case 3:

Different Plastics, Hard to Differentiate
(PE and PVC on Cellulose Filter)



2D+SNV

Challenge 3a: ??

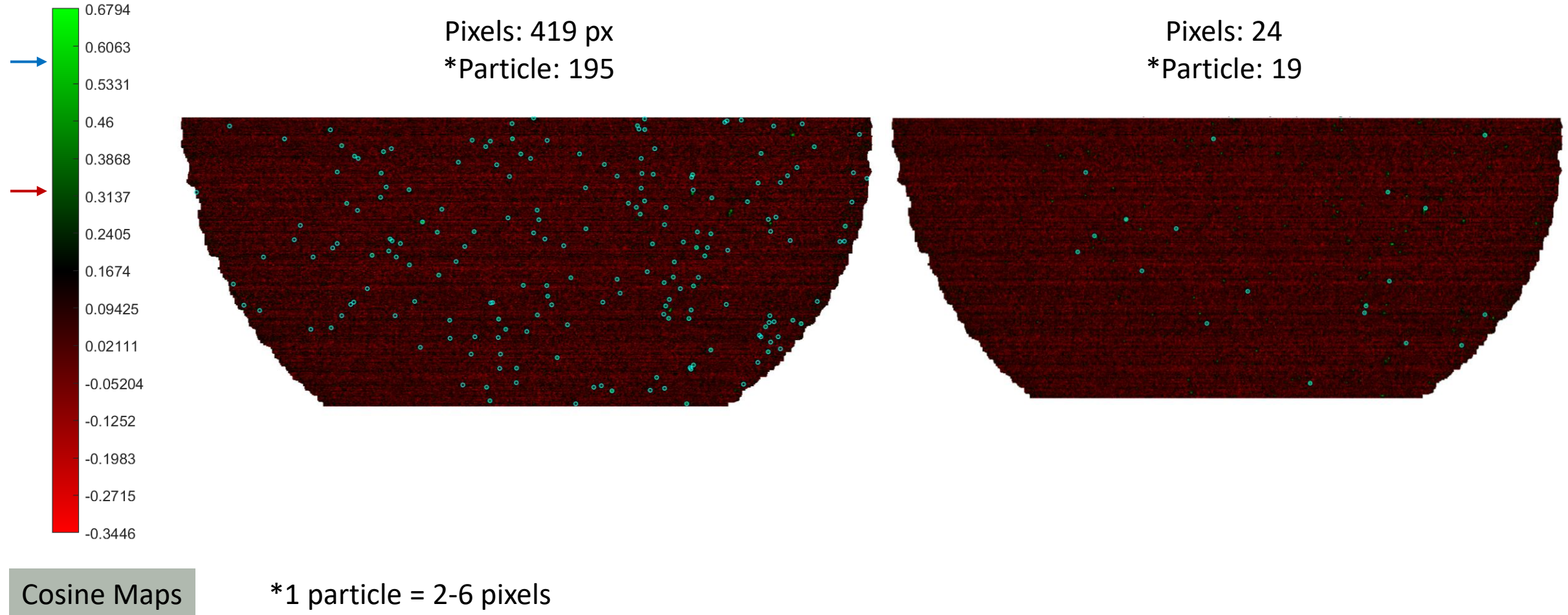


Cosine Maps

*1 particle = 2-6 pixels

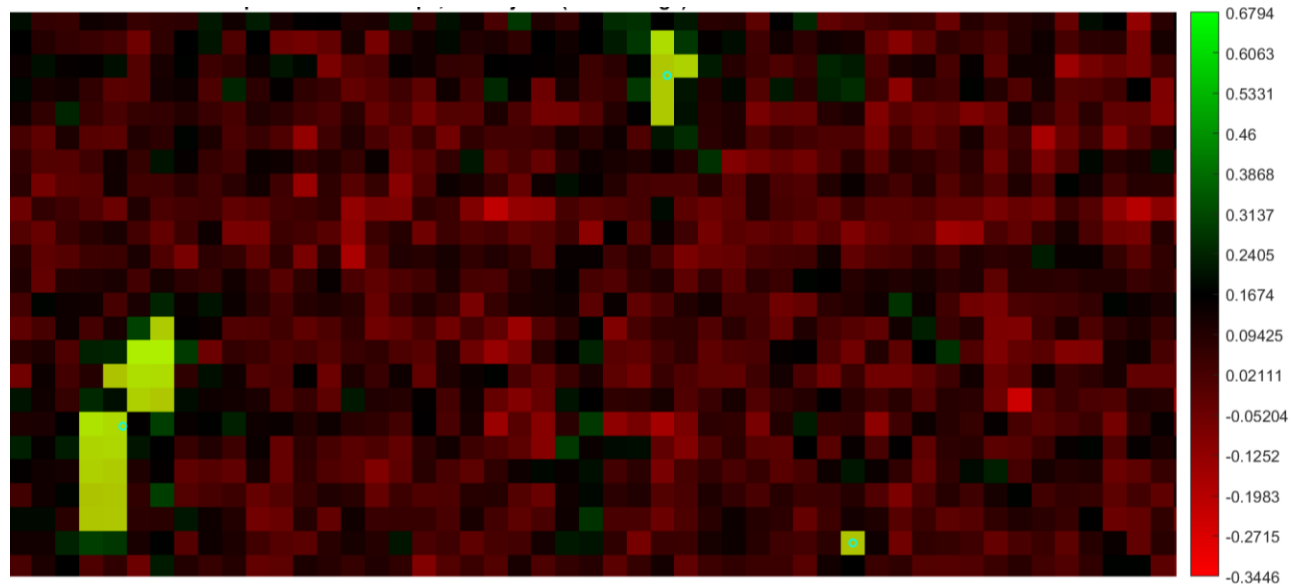
Challenge 3a: Image Segmentation - Setting the Cutoff

- Are we including noise?

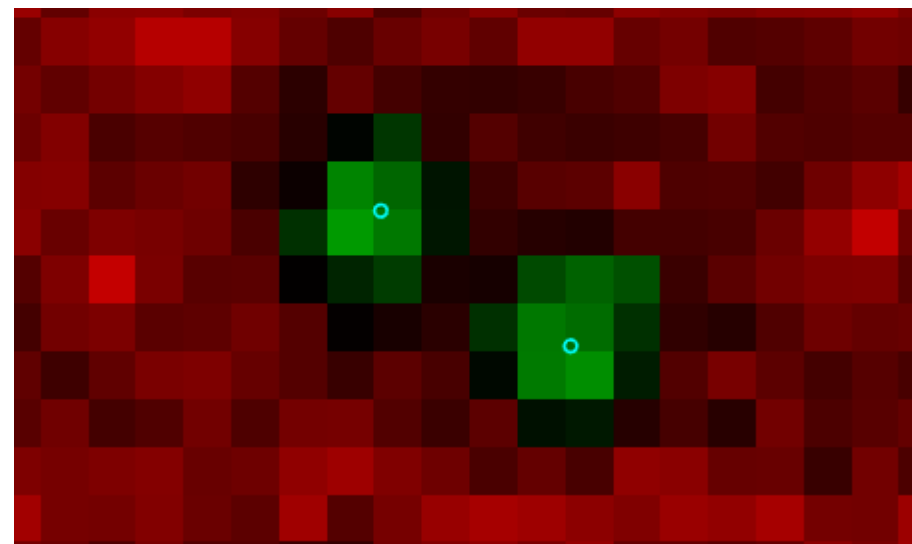


Challenge 3b: ??

PE



PVC

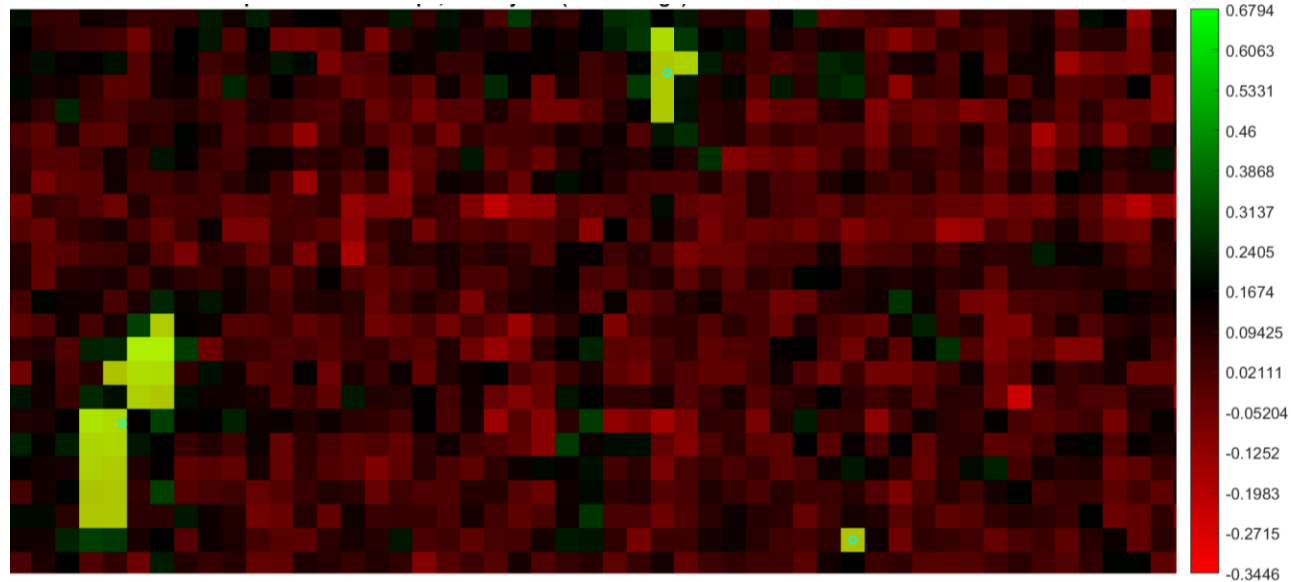


The pixel size is not a real particle size

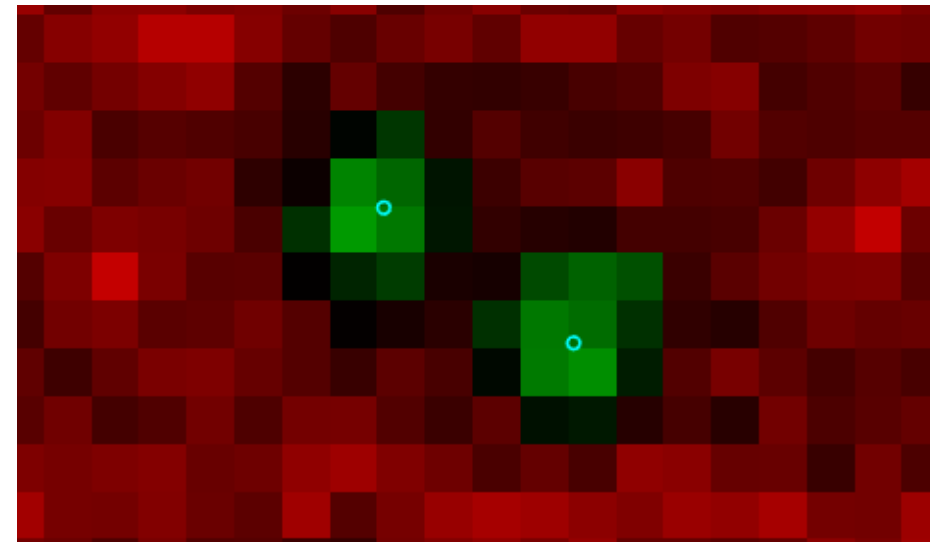
Challenge 3b: Particle Counting

- What is a particle?

PE



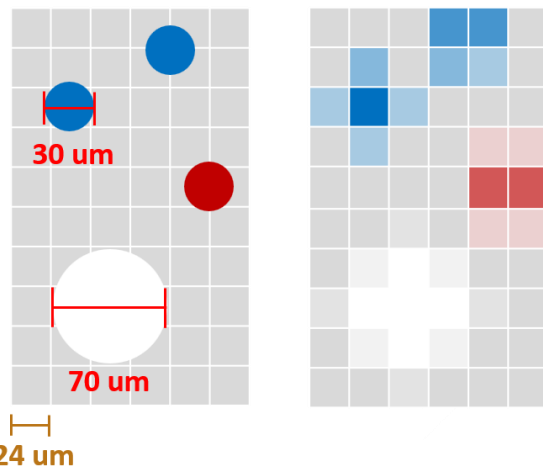
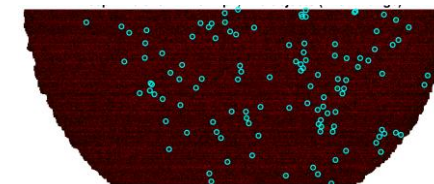
PVC



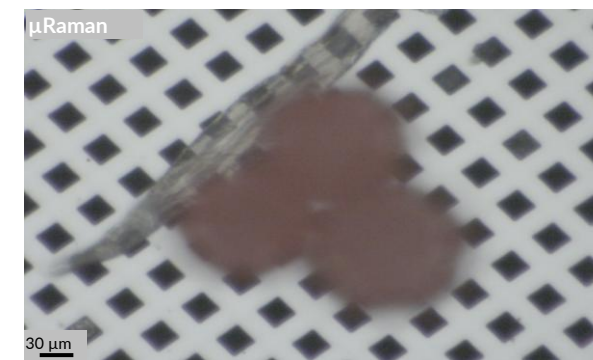
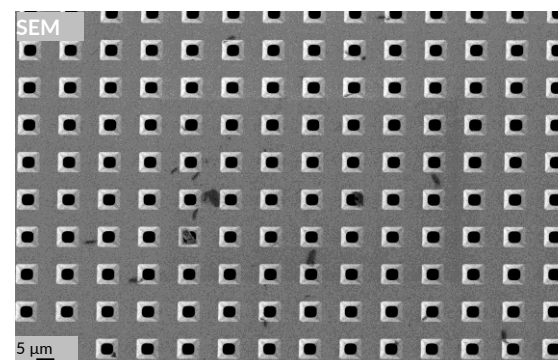
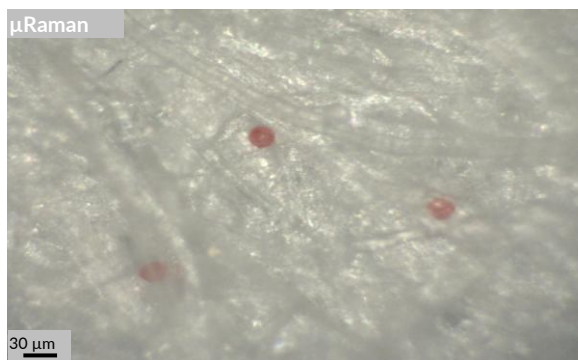
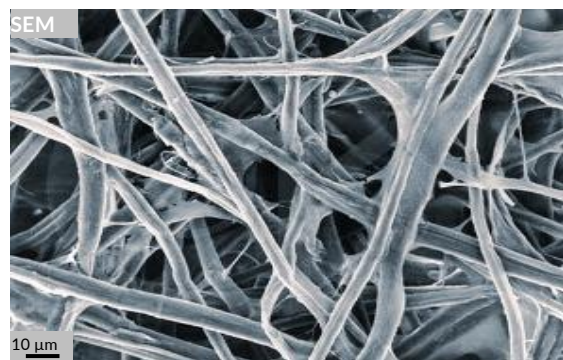
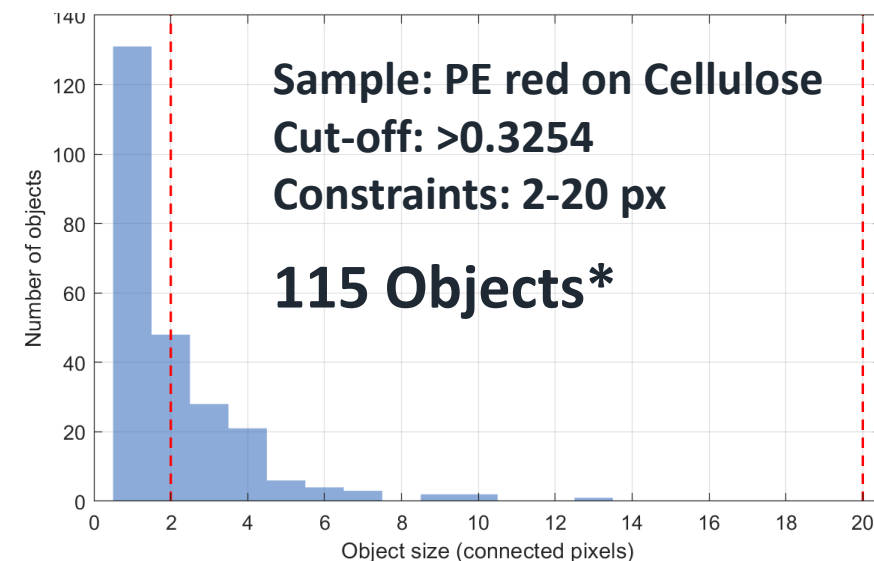
The pixel size is not a real particle size

Perspective: On Setting a Cut-off and Counting a Particle

- Make an informed decision based on the pixel limit and particle size
- Report counts as a proxy for the number of particles only
- Unknown: estimate, report limits



Example, arbitrary cut-off		Cellulose	Silicon
		Min # Pixels	Min # Pixels
PE	0.5?	1?	2?
PVC	0.5?	2?	5?



Summary of decisions and uncertainties

Major decisions

- ❖ Reference spectra
- ❖ Preprocessing
- ❖ Detection Threshold
- ❖ Particle size (# pixels)

depends on



Sample

Filter

Plastic type

Particle Size

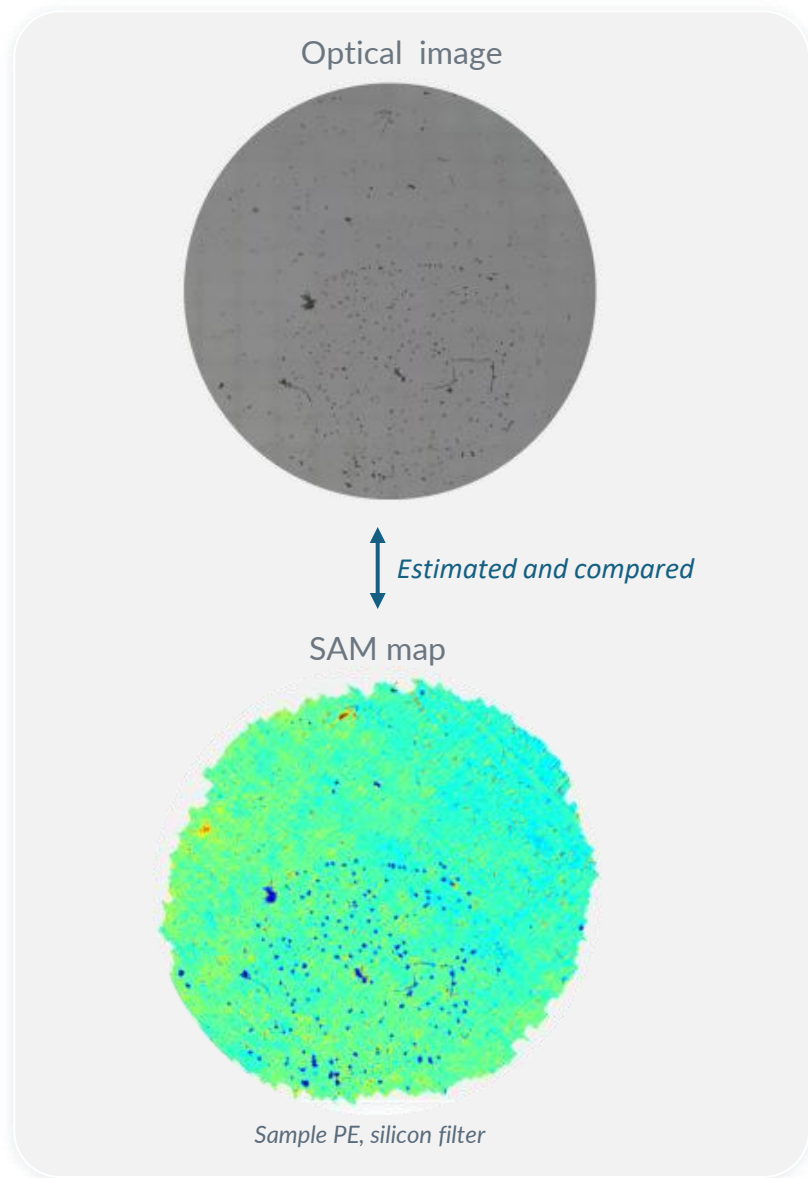
Resolution (pixel size)

The trap

We tuned the pipeline using the answer.

In **real analysis**, the answer is what we are trying to find.

Perspective: On Final Reported Detections



- Compare counts with standard methods (RAMAN/FTIR)

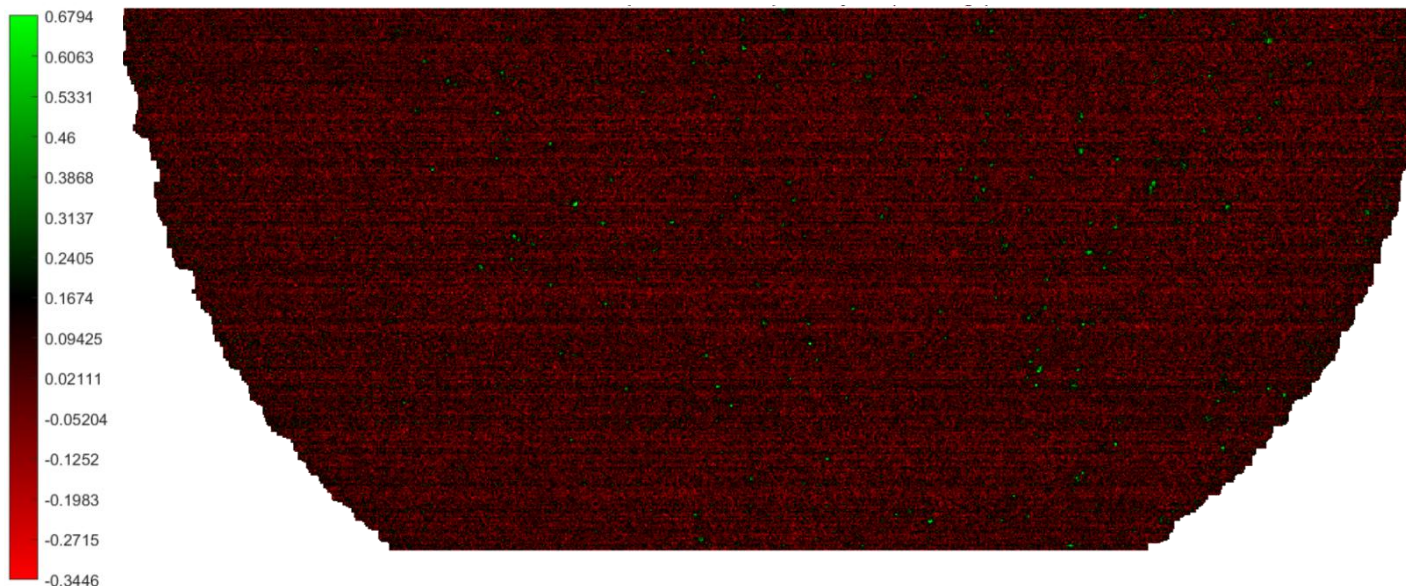
$$\text{recovery \%} = 100 \times \frac{\text{HSI counted objects}}{\mu\text{Raman Optical Image counted particles}}$$

	in Silicon	in Cellulose
blue PE	78 ± 15 %	85 ± 16 %
blue PE	98 ± 42 %	134 ± 14 %
PVC	77 ± 52 %	110 ± 38 %
Mix	92 ± 24 %	105 ± 24 %

- No way of checking if the pixel we counted was the exact spatial location of the particle in the optical image

We can detect, but are we sure **we are counting the correct pixels?**

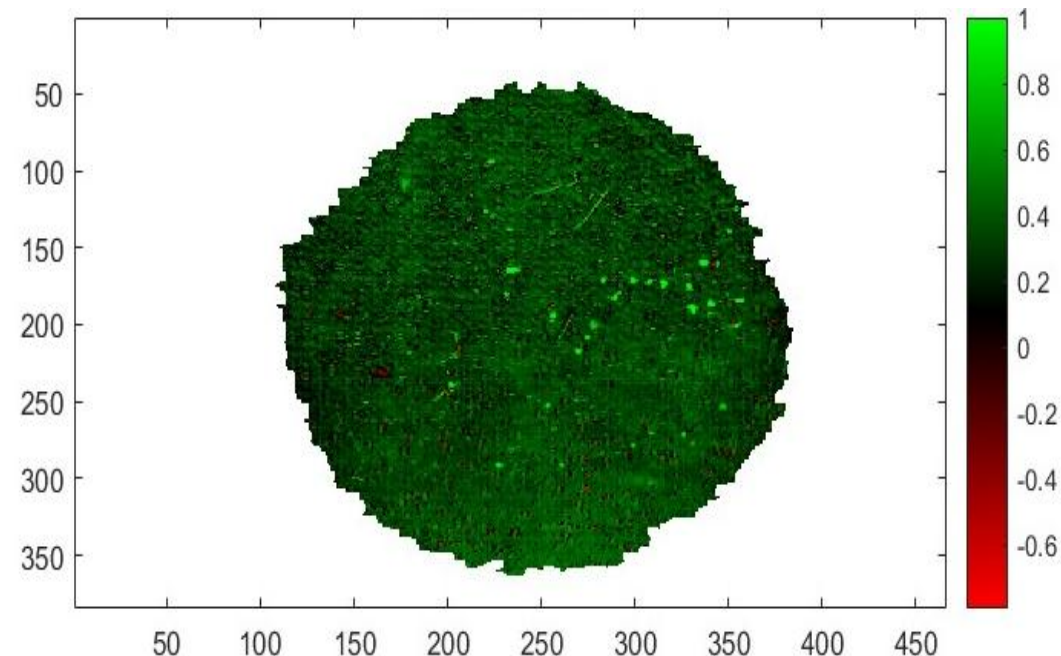
PE red on Cellulose Filter Cosine Map against reference



Perspective:

Framework for validation
needed, not more proof of
detection...

PE red on Silicon Filter Cosine Map



How? Spiked recovery? RAMAN/FTIR?

What do *you* think?

Presentation Acknowledgement



This work is presented with the support of **COST Action CA23131 ICPLASTIC** (ISO compatible, efficient and reproducible protocols/equipment for mICro-nanoPLASTIC detection through machine-learning), supported by COST (European Cooperation in Science and Technology).



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